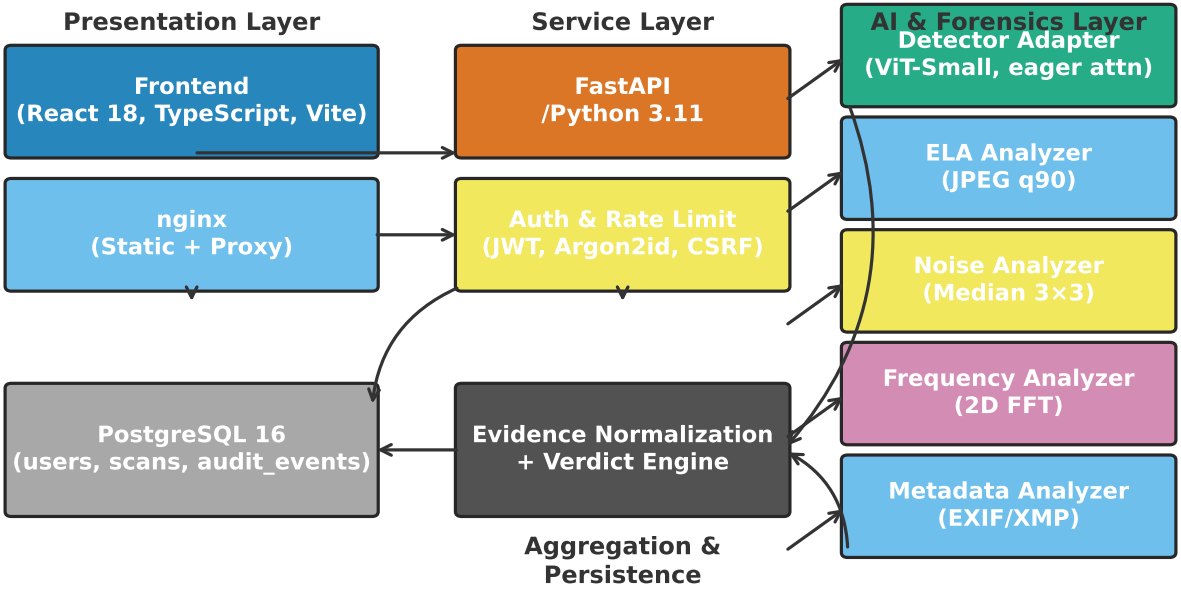
Presentation Layer
Service Layer
AI & Forensics Layer
Frontend
(React 18, TypeScript, Vite)
FastAPI
/Python 3.11
Detector Adapter
(ViT-Small, eager attn)
ELA Analyzer
(JPEG q90)
nginx
(Static + Proxy)
Auth & Rate Limit
(JWT, Argon2id, CSRF)
Noise Analyzer
(Median 3×3)
Frequency Analyzer
(2D FFT)
PostgreSQL 16
(users, scans, audit_events)
Evidence Normalization
+ Verdict Engine
Metadata Analyzer
(EXIF/XMP)
Aggregation &
Persistence